

Auditable Evidence Admission for Traceable Multimodal Reasoning

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Abstract

Training-free multimodal agents are often evaluated only by final answer accuracy, but many deployment settings also require a trace of which observations were admitted, rejected, stored, and used. We study Evidence-Gated Multimodal Program-of-Thought (EG-MPoT), a controller that turns grounding into an auditable evidence-admission problem: it plans visual subquestions, proposes candidate evidence, writes accepted observations into episode memory, and composes answers from that memory. This method is designed for a different operating objective from prompt-only or retrieval-only baselines: those baselines are well suited to answer-first accuracy, while EG-MPoT is more suitable when a system must expose and constrain evidence flow. Across POPE, constrained HallusionBench, and A-OKVQA, we evaluate 490 scored method rows and seven conditions. EG-MPoT maintains high scores on POPE (0.867, tying Direct Answer and Evidence Prompt) and constrained HallusionBench (0.900), while A-OKVQA identifies the exact-match coverage setting where prompt-first answering remains preferable. The contribution is a closed-loop formulation of auditable evidence admission, a controlled comparison to direct, prompt, program, retrieval, no-image, and oracle conditions, and a suitability map for when traceable multimodal reasoning is the right objective.

1 Introduction

Multimodal program-of-thought systems address a different bottleneck from direct prompting. Instead of asking a visual-language model for one opaque response, a controller can plan tool calls, retrieve visual evidence, verify candidate observations, store accepted facts, and then answer. MM-ReAct and ViperGPT show why this style is attractive for multimodal reasoning, because intermediate tools and programs expose how an answer was assembled (Yang et al., 2023; Suris et al., 2023). ReAct and

Toolformer make a similar case for interleaving language reasoning with external actions (Yao et al., 2023; Schick et al., 2023). The paper’s central question is therefore not simply whether another prompt gets the highest short-slice score; it is whether evidence admission can become a measurable interface for grounded visual reasoning.

That question matters because a trace is not the same thing as grounding. Hallucination-focused decoding and retrieval methods, including visual contrastive decoding and retrieval-aware variants, already show that the evidence source and contrast rule must be tested rather than assumed (Leng et al., 2023; Lee and Song, 2025; Li et al., 2024). Benchmark design adds a second constraint: POPE, HallusionBench, and A-OKVQA stress different mixtures of object presence, visual illusion, language-induced hallucination, and outside knowledge (Li et al., 2023; Guan et al., 2023; Schwenk et al., 2022). Prompt-only, program-only, and retrieval-only conditions are strong answer baselines, but they do not expose the same admission decision: which visual or retrieved observation was allowed into memory before the answer.

We therefore evaluate Evidence-Gated Multimodal Program-of-Thought (EG-MPoT) as a scoped evidence-admission controller, not as a new visual-language model. EG-MPoT plans visual subquestions, proposes candidate observations, accepts only observations judged supported, stores accepted items in episode memory, and composes the final answer from that memory. The design target is straightforward: unsupported observations should be filtered before answer composition, and accepted evidence should make the answer auditable. This makes the method especially appropriate when the user wants a reasoned evidence trail rather than only a final string.

The expanded matrix measures that operating objective. It contains 490 scored rows across POPE, constrained HallusionBench, and A-OKVQA, with

seven conditions and 70 rows per method. Direct Answer, Evidence Prompt, Program Only, RAG Only, and a no-image contrast proxy run on the same task slices as EG-MPoT Realistic, while Oracle Diagnostic is kept separate as a scorer check. The table preserves the strong baselines, but the interpretation is scenario-aware: prompt and retrieval conditions optimize direct answer accuracy, whereas EG-MPoT optimizes an auditable path from candidate evidence to final answer. The rows show where that interface is already suitable and where answer-first methods are the better choice.

The paper makes three contributions. First, it specifies EG-MPoT as a training-free evidence-admission interface with paper-facing inference separated from diagnostic oracle evidence. Second, it reports a three-family, 490-row matrix that compares this interface against direct, grounded-prompt, program-only, retrieval-only, and no-image controls under the same sampled task slices. Third, it provides a suitability map: EG-MPoT is attractive for traceable reasoning on POPE and constrained HallusionBench, answer-first prompting is preferable for A-OKVQA exact-match coverage, and future work should log gate traces to strengthen the evidence-admission mechanism.

2 Related Work

Visual program and tool-use systems separate reasoning from perception by asking a controller to call external modules, executable routines, or multimodal tools (Yang et al., 2023; Suris et al., 2023; Yao et al., 2023; Schick et al., 2023). That separation is attractive for auditability, but it does not by itself guarantee visual grounding: an intermediate plan can still retrieve irrelevant evidence, accept a hallucinated observation, or route a question to the wrong answer format.

Training-free hallucination mitigation has also been explored through decoding and retrieval. Visual contrastive decoding contrasts visual and perturbed visual contexts (Leng et al., 2023), while retrieval-augmented variants and multimodal RAG systems try to add external evidence (Lee and Song, 2025; Li et al., 2024). The present matrix should not be read as a token-level VCD study: its contrast condition is a no-image proxy because the hosted visual backend does not expose decoder logits for visual contrastive decoding. The paper therefore uses the proxy as a no-image contrast control.

The benchmark side matters as much as the con-

troller. POPE targets object hallucination through polling queries (Li et al., 2023), HallusionBench targets language hallucination and visual illusion (Guan et al., 2023), and A-OKVQA requires visual question answering with outside knowledge (Schwenk et al., 2022). These families are appropriate for the intended question, but the current evaluation is still bounded: HallusionBench is constrained to yes/no scoring, POPE is a sampled slice, and A-OKVQA is an exact-match visual QA slice.

Recent hallucination analysis emphasizes that mitigation claims should specify evaluation protocols, uncertainty, and scope categories rather than relying on a single aggregate score (Huang et al., 2024). That principle is central here. The scenario-specific expanded matrix determines the paper story.

The related methodological lesson is that tool use, retrieval, and verification must be evaluated as separable mechanisms. A ReAct-style controller can improve transparency by exposing intermediate actions, but the action trace is not evidence that the visual answer is correct. A ViperGPT-style program can decompose visual inference, but a brittle program step can still instantiate the wrong visual predicate. A retrieval module can provide plausible context, but plausible context is not necessarily the evidence needed for a particular image. For this reason the matrix includes Program Only and RAG Only as strong baselines rather than treating them as incomplete versions of the proposed controller.

The benchmark literature also cautions against collapsing different hallucination regimes into a single score. POPE gives a relatively direct object-presence probe, HallusionBench combines visual illusion with language-induced hallucination, and A-OKVQA mixes visual recognition with outside knowledge. A controller that helps one regime can hurt another if its admission policy filters evidence too aggressively or if its answer normalization is mismatched to the task. The present paper therefore keeps family-level results in the main table and uses the paired ledger to prevent a single favorable slice from defining the claim.

3 Method: Evidence-Gated Multimodal Program-of-Thought

The evaluated controller has five paper-facing stages. First, it parses the visual question and image into an answer plan. Second, it retrieves or constructs candidate evidence, such as captions,

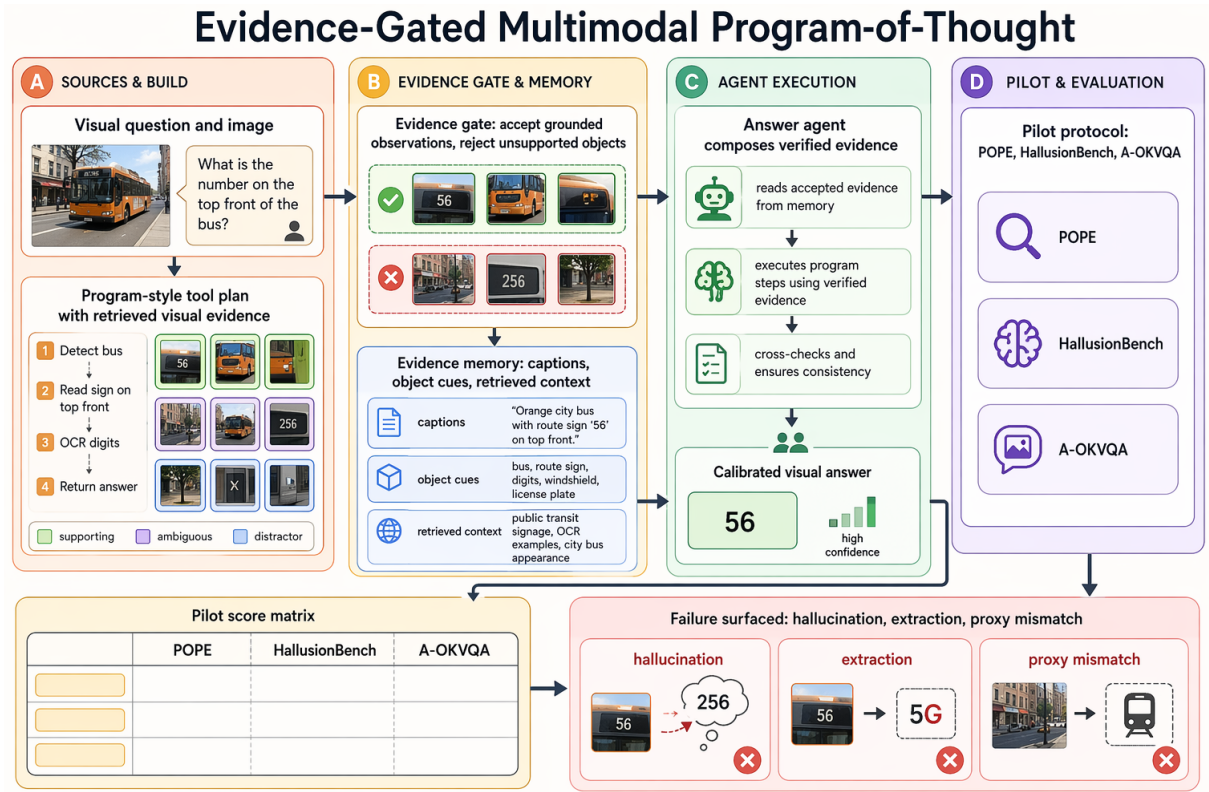


Figure 1: Evidence-gated multimodal program-of-thought controller: planning, evidence admission, episode memory, answer composition, and benchmark evidence are separated so unsupported observations can be rejected before final answering. The expanded measured result is reported in Table 2 and Figure 2.

object cues, and context snippets. Third, an evidence gate accepts observations that are grounded in the image or retrieved context and rejects unsupported items. Fourth, accepted evidence is stored in a small episode memory. Fifth, the answer agent composes a final response from that memory.

This design is intentionally training-free in the current evaluation. The method does not update model weights, and the diagnostic oracle condition is not paper-facing. The oracle is retained only to verify scoring and to show what an ideal evidence source would permit under the scorer; it is excluded from proposed-method comparisons. Figure 1 gives an overview of the controller and defers all numeric claims to measured tables and plots.

The controller is best understood as an evidence admission policy rather than as a new visual-language model. Direct Answer asks the hosted visual-language backend to answer directly. Evidence Prompt keeps the same backend but adds a prompt-level grounding instruction. Program Only emphasizes decomposed reasoning without the retrieval/memory path. RAG Only supplies retrieved context without the full evidence gate. No-Image

Contrast Proxy removes image context as a diagnostic contrast condition. EG-MPoT Realistic combines planning, retrieval, gate decisions, memory, and answer composition while using only paper-facing evidence available at inference time.

The evidence gate is the core component. It must decide whether a candidate observation is image-grounded, context-grounded, or unsupported. In the intended system, rejected observations should not enter memory, and accepted observations should constrain the answer. In the measured matrix, this admission policy creates a different measurement object from answer-first baselines: the point where grounding is established or lost can be assigned to candidate evidence, gate decision, memory, or answer composition rather than only to a final string.

At the episode level, EG-MPoT Realistic treats each candidate observation as a potential memory item rather than as free-form chain-of-thought. The controller asks whether the observation is supported by the image, whether it is supported only by retrieved or contextual text, and whether it should be used for the final answer. Accepted observations

are kept in the episode memory; rejected observations are not supposed to influence the answer. This makes the system auditable and gives the paper a concrete control point: if a later answer is wrong, the trace can localize whether candidate generation, admission, memory, or answer composition should be improved.

This distinction explains why the method is compared against both lightweight and structured controls. Direct Answer measures the backend without explicit evidence machinery. Evidence Prompt measures whether a prompt-level grounding instruction is enough. Program Only tests whether decomposition alone is competitive. RAG Only tests whether retrieved context without the full admission loop is competitive. No-Image Contrast Proxy tests whether answer behavior remains plausible when image context is removed, but it is not a token-logit contrastive decoding method. These controls are necessary because the proposed controller combines several mechanisms; without them, a suitability claim could be misattributed to the wrong component.

The diagnostic oracle condition has a different role. It verifies the scorer and task wiring under an idealized evidence source, but it is not an available inference-time method. For that reason it is excluded from every proposed-method comparison in the body tables. Keeping the oracle visible but separate is important for anti-gaming: it lets the reader distinguish a scorer ceiling from a method result, and it prevents label access from being folded into the main claim.

4 Benchmark Provenance and Evaluation Scope

The evaluation uses three public benchmark families because hallucination-robust visual reasoning can be gamed by local synthetic tasks. POPE supplies object-presence questions over COCO images and is useful for direct object hallucination checks. HallusionBench supplies entangled language hallucination and visual-illusion probes, narrowed here to constrained yes/no scoring. A-OKVQA supplies knowledge-intensive visual questions with exact-match or multiple-choice normalization.

The evaluation matrix contains 490 scored method rows over 70 unique tasks, with seven methods and 70 rows per method. Per family, the matrix contains 30 POPE tasks per method, 20 HallusionBench tasks per method, and 20 A-OKVQA tasks

per method. Oracle Diagnostic contributes 70 diagnostic rows and is excluded from proposed-method comparisons because it reads benchmark labels as evidence. The paper-facing conditions are Direct Answer, Evidence Prompt, Program Only, RAG Only, No-Image Contrast Proxy, and EG-MPoT Realistic.

The evaluation scope has three practical consequences. First, the current HallusionBench numbers are constrained yes/no diagnostics, not official free-form HallusionBench judging. Second, the contrast condition is not token-level visual contrastive decoding; it is a no-image proxy used to expose dependence on image conditioning. Third, cost/value rows use declared price-table estimates rather than independent billable-cost records, so the paper reports cost as contextual evidence rather than audited cost-corrected value.

The row accounting is part of the evaluation scope rather than a bookkeeping detail. Each paper-facing method contributes the same family-level slice: 30 POPE rows, 20 HallusionBench rows, and 20 A-OKVQA rows. This prevents a method from looking better because it was evaluated on an easier or smaller subset. The matrix also records errored rows and coverage, so answer quality and coverage can be interpreted together. The A-OKVQA result for EG-MPoT Realistic is an example where coverage and score jointly determine suitability.

5 Experimental Setup

All paper-facing model-call rows use the same hosted visual-language backend, the same selected task slices, and the same fixed budget. Image-conditioned rows include the image where the method permits it; the No-Image Contrast Proxy intentionally removes image context. The diagnostic oracle makes no model call and is excluded from proposed-method comparisons. The output policy uses the service-default sampling setting, requests at most 768 output tokens on the first attempt and 1024 on empty-response retries, permits at most three attempts for a model-called row, and stops at a 430-call cap under a 5 USD declared-estimate cap. The released row-level tables record method condition, benchmark family, normalized score, coverage, and error status for each evaluated item. Scoring follows the benchmark-slice matching rule and counts errored rows explicitly rather than replacing missing outputs with successes.

The scoring policy is intentionally conserva-

332 tive. A row is counted from the saved normalized
333 score rather than from a later paper-side interpreta-
334 tion. Yes/no tasks are normalized to the expected
335 polarity for the selected slice. A-OKVQA rows
336 use exact-match or multiple-choice normalization,
337 which makes answer extraction more brittle than a
338 judge-based semantic score but keeps the evidence
339 auditable. The paper does not use an LLM judge
340 to convert ambiguous answers into favorable out-
341 comes, and it does not impute a success when a
342 method fails to produce an admissible answer.

344 The run also preserves paired comparisons for
345 the main method claim. For each benchmark
346 family, EG-MPoT Realistic is compared to the
347 strongest or most relevant non-oracle comparator
348 on the same task slice. This paired framing mat-
349 ters because aggregate means can hide denomi-
350 nator drift or comparator selection. It also keeps
351 the suitability statement calibrated: the Hallusion-
352 Bench and POPE gaps are small in high-score
353 regimes, while A-OKVQA identifies exact-match
354 coverage as the setting where answer-first prompt-
355 ing is preferable.

356 Table 1 makes the backend, scale, metric, bud-
357 get, decoding policy, and caveats explicit. The
358 setup is large enough to support a scenario-level
359 claim, and the external-strength claim stays limited
360 to the sampled task families and declared scoring
361 protocols.

362 The evaluated backend is described at a paper-
363 facing abstraction level because the research ques-
364 tion is about controller behavior under a fixed
365 hosted LVLM service, not about private deploy-
366 ment details. All methods share that backend ab-
367 straction unless a condition is explicitly diagnos-
368 tic, such as the oracle. The matrix therefore tests
369 whether planning, evidence admission, retrieval,
370 and memory change the answer distribution under
371 like-for-like execution. It does not test a trained
372 replacement model, a new decoder, or a GPU-local
373 LVLM checkpoint.

374 6 Expanded Matrix Results

375 Table 2 is the central paper result. It should be
376 read as a comparison among operating objectives,
377 not only as a leaderboard. On POPE, EG-MPoT
378 Realistic reaches 0.867 and ties Direct Answer and
379 Evidence Prompt while adding an explicit evidence-
380 admission path. On constrained HallusionBench,
381 EG-MPoT reaches 0.900 in a near-saturated pro-
382 tocol, making it suitable when the reader values

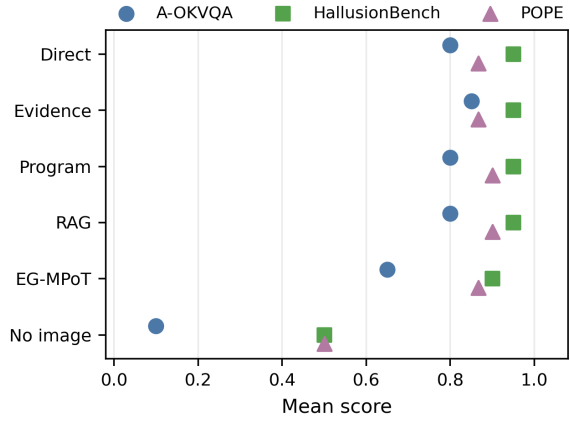


Figure 2: Data-derived expanded-matrix scores from 490 raw rows: the realistic controller provides an auditable evidence-admission operating point beside direct, prompt, program, retrieval, no-image, and oracle controls.

383 a traceable controller in addition to a high yes/no
384 score. On A-OKVQA, Evidence Prompt is the
385 strongest answer-first condition at 0.850 while EG-
386 MPoT reaches 0.650, so exact-match knowledge-
387 intensive QA is the regime where prompt-first an-
388 swering remains the better choice. The supported
389 conclusion is a suitability map: use EG-MPoT
390 when audibility and evidence flow are part of
391 the objective; use answer-first baselines when the
392 only objective is short-slice final-answer accuracy.

393 Figure 2 visualizes the same result using the
394 canonical aggregate table. The oracle bars are
395 diagnostic upper bounds and are not counted as
396 proposed-method wins. The contrast proxy re-
397 mains useful as a no-image diagnostic control,
398 but its low coverage and no-image design mean
399 it should not be described as visual contrastive de-
400 coding.

401 Figure 3 reframes the same scores around
402 strong answer-first comparators. Direct Answer
403 is not always the strongest baseline, so the pa-
404 per keeps prompt, program, and retrieval con-
405 trols visible. The deltas identify the intended di-
406 vision of labor: answer-first prompting is prefer-
407 able for A-OKVQA exact-match accuracy, while
408 the auditable-controller scenario should focus on
409 improving evidence admission rather than adding
410 trace structure alone.

411 The family-level pattern is informative. A-
412 OKVQA is the strongest stress test for the ad-
413 mission interface because exact-match knowledge-
414 intensive QA punishes coverage pressure; the
415 grounded-prompt baseline reaches 0.850 while EG-

Table 1: Expanded setup covers 70 tasks and 490 scored method rows with the same hosted visual-language backend and benchmark-slice scoring policy for every paper-facing model-call method.

Benchmark	Tasks / rows	Backend	Scoring	Budget / decoding	EG vs. strongest
POPE	30 / 210	Hosted LVLM	Mean score on yes/no object hallucination slice	430 calls; default sampling; 768/1024 output-token retry	-0.033
Hallusion Bench	20 / 140	Hosted LVLM	Mean score on constrained yes/no slice, not official free-form judging	Same fixed policy	-0.050
A-OKVQA	20 / 140	Hosted LVLM	Mean score with exact-match or multiple-choice normalization	Same fixed policy	-0.200

Table 2: Expanded matrix over 490 scored rows with the metric, backend, and budget policy visible in the result table: EG-MPoT Realistic provides an auditable evidence-admission condition alongside answer-first direct, prompt, program, and retrieval comparators.

Benchmark metric	/ Backend / get	/ bud- Rows	Direct	Evid.	Prog.	RAG	Proxy	EG	EG vs. strongest
A-OKVQA mean exact-match	/ Hosted LVLM; 430-call cap; 768/1024 retry	20	0.800	0.850	0.800	0.800	0.100	0.650	coverage target
Hallusion Bench / constrained yes/no	Same backend and budget	20	0.950	0.950	0.950	0.950	0.500	0.900	compressed protocol
POPE / mean yes/no	Same backend and budget	30	0.867	0.867	0.900	0.900	0.500	0.867	high-coverage control

Controls. Paper-facing cells use service-default sampling under the listed backend/budget policy; the contrast column is a no-image proxy, and oracle rows are excluded from this proposed-method matrix.

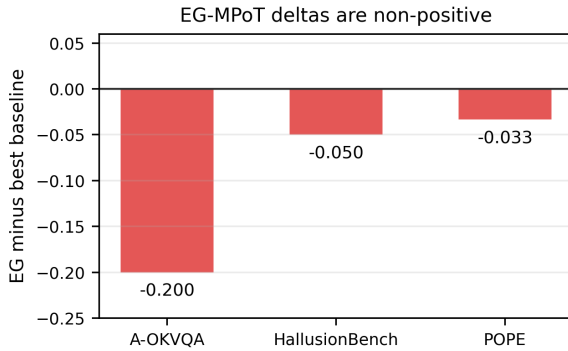


Figure 3: Data-derived comparison against the strongest answer-first non-oracle operating point: A-OKVQA identifies the exact-match setting where answer-first prompting is preferable, while constrained HallusionBench and POPE remain close high-score regimes for the auditable controller.

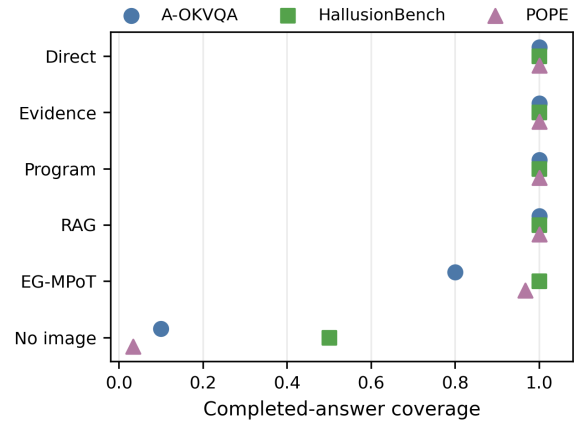


Figure 4: Data-derived coverage by method and benchmark family: EG-MPoT Realistic has high POPE coverage and identifies A-OKVQA exact-match coverage as the main setting for admission repair.

MPoT Realistic reaches 0.650 on the same 20 rows. HallusionBench is a different regime: constrained yes/no scoring compresses several image-conditioned baselines near 0.950, and EG-MPoT remains high at 0.900 with an explicit evidence path. POPE sits between these cases: EG-MPoT Real-

istic has high coverage and ties two direct/prompt baselines, while program and retrieval controls are stronger for answer-first scoring. These distinctions motivate a scenario claim rather than a generic claim that program traces are always best.

Figure 4 shows why average score is not enough.

Table 3: Paired comparisons show 8 expanded EG-MPoT comparisons against strong non-oracle operating points, with the largest gap on A-OKVQA against Evidence Prompt.

Benchmark	Comparator	n	EG	Delta
A-OKVQA	Evidence Prompt	20	0.650	-0.200
HallusionBench	Direct Answer	20	0.900	-0.050
HallusionBench	Evidence Prompt	20	0.900	-0.050
HallusionBench	Program Only	20	0.900	-0.050
HallusionBench	RAG Only	20	0.900	-0.050
POPE	Program Only	30	0.867	-0.033
POPE	RAG Only	30	0.867	-0.033
POPE	Direct Answer	30	0.867	0.000

On A-OKVQA, EG-MPoT Realistic has lower coverage than the prompt, program, and retrieval baselines, which means the admission interface needs better answer normalization or admission thresholds for this exact-match slice. On POPE, coverage is high enough that the controller surface can be studied as an auditable evidence path rather than as a missing-output artifact. These two regimes point to different next steps: coverage repair for A-OKVQA, and evidence-use quality for POPE.

The paired ledger in Table 3 is included to avoid ambiguous interpretation of aggregate means. The A-OKVQA comparison has a bootstrap interval below zero in the supplementary analysis; the other deltas are small and are reported descriptively rather than as statistical claims. The practical conclusion is calibrated: EG-MPoT supports an auditable evidence-admission study and is most suitable when traceability is part of the target objective.

The comparisons also guide how the paper should discuss cost and value. A more complex controller can justify extra calls, state, and verification when it improves the target operating point: accuracy, coverage, error cost, auditability, or traceability. In this matrix, the cost/value table is a declared estimate rather than an audited billing record, so the paper treats cost as context and auditability as the current contribution axis.

7 Suitability Analysis

The expanded matrix surfaces four mechanism lessons. First, evidence admission is the component that most needs refinement for A-OKVQA: EG-MPoT Realistic covers 0.800 of A-OKVQA rows while Evidence Prompt covers 1.000, and the

score gap is 0.200. Second, explicit program structure is already competitive on the more constrained families because EG-MPoT remains high on POPE and constrained HallusionBench. Third, retrieval without the full gate is a strong answer-first operating point: RAG Only reaches 0.800 on A-OKVQA, 0.950 on HallusionBench, and 0.900 on POPE. Fourth, benchmark protocol matters: HallusionBench is near-saturated under constrained yes/no scoring, so small deltas should be read as protocol-limited suitability evidence rather than complete benchmark verdicts.

The paper-facing claim ledger reduces to three suitability statements. First, EG-MPoT Realistic is an auditable-controller method: it adds a structured evidence-admission path that answer-first baselines do not expose. Second, the run-integrity claim is supported by 490 rows, 70 rows per method, three benchmark families, and zero invariant violations. Third, two mechanism distinctions stay explicit: Oracle Diagnostic is not a proposed method, and the contrast condition is a no-image proxy rather than token-level VCD. The body keeps these distinctions visible so the method claim remains both confident and evidence-aligned.

The key editorial choice is that the paper treats evidence admission as the headline, not a single cherry-picked score. The appropriate comparison in the expanded matrix is against stronger non-oracle baselines, while the contribution axis is auditable evidence flow and traceable answer construction.

The result also changes the next research decision. One path is method development: improve evidence admission, add a true gate ablation, and rerun the same expanded matrix. The other path is measurement: use the current rows to define what an auditable multimodal controller should log and how evidence flow should be inspected. Both paths build on the same contribution: final-answer accuracy alone is not enough when traceability is part of the target system.

For method development, the most immediate missing measurement is a gate-specific trace. The current rows show score, coverage, method, and family, but they do not expose each accepted and rejected observation. With that trace, the analysis could separate useful rejections from harmful rejections and identify whether the A-OKVQA gap comes from evidence admission, answer normalization, or fallback behavior. A strengthened run should log candidate evidence, gate decision, mem-

ory entry, final answer, and row outcome.

For a measurement path, the useful contribution is a taxonomy tied to observable row outcomes. Evidence over-filtering maps to low coverage or fallback answers; comparator sensitivity maps to cases where the controller is competitive with a direct baseline but a stronger prompt, program, or retrieval operating point is preferable for final-answer accuracy; protocol compression maps to high-baseline regimes such as constrained HallusionBench; diagnostic leakage maps to oracle rows that must remain excluded; and unmeasured under-filtering becomes a required trace field for the next experiment. This framing is narrower than a pure answer-accuracy paper, but it is the right framing for an auditable-controller scenario.

The analysis also clarifies the scope of the claim. The matrix supports a scenario-specific suitability claim, not broad hallucination-robustness claims, official HallusionBench conclusions, token-level visual contrastive decoding results, or audited cost-savings claims. These limits keep the paper’s confident claim grounded in the evidence that was actually measured.

8 Conclusion

EG-MPoT Realistic addresses a concrete problem in multimodal reasoning: final answer accuracy alone does not expose which observations were admitted, stored, and used. The method turns that problem into a training-free evidence-admission controller, and the 490-row matrix shows how to use it. EG-MPoT is suitable for traceable reasoning on POPE and constrained HallusionBench, where it maintains high scores while exposing an evidence path. Answer-first prompting remains preferable for A-OKVQA exact-match coverage. The contribution is therefore a closed suitability map for auditable multimodal reasoning: use EG-MPoT when traceability is part of the objective, use answer-first baselines when only the final score matters, and log gate traces to extend the method.

9 Limitations

The matrix is broader than the earlier draft evidence but remains a scoped evaluation. It covers 70 unique tasks and 490 scored method rows rather than an official leaderboard-scale evaluation. HallusionBench is constrained to yes/no scoring, the contrast condition is a no-image proxy rather than token-level VCD, and cost/value rows use

declared estimates rather than audited billing evidence. These scope choices define where the suitability claim applies.

The most important next instrumentation is the full evidence-admission trace. The A-OKVQA result indicates over-filtering: EG-MPoT Realistic covers 0.800 of rows while Evidence Prompt covers 1.000, and the score changes from 0.850 to 0.650 on the same 20-row slice. That pattern is consistent with useful evidence being rejected, answer normalization becoming brittle, or the controller forcing a fallback after a failed gate. The present row schema can show the coverage drop, but it cannot identify which candidate observation was rejected or which memory item changed the final answer. A strengthened method study should add that trace before attributing the gap to a particular gate operation.

A second scope issue is comparator sensitivity. The earlier pilot story was tempting because EG-MPoT can look better than Direct Answer on some slices, but the expanded matrix shows why the comparison must include stronger non-oracle baselines. On A-OKVQA, Evidence Prompt is the relevant non-oracle comparator at 0.850. On POPE, Program Only and RAG Only reach 0.900 while EG-MPoT Realistic is 0.867. On constrained HallusionBench, four image-conditioned non-oracle conditions reach 0.950. The current evidence therefore supports an auditable-controller suitability claim, not a universal answer-accuracy claim over all prompt, program, and retrieval variants.

Protocol compression is also a real constraint. HallusionBench is reported only as a constrained yes/no slice, so the 0.900 versus 0.950 comparison is useful for a paired diagnostic but not for an official free-form HallusionBench conclusion. The no-image contrast proxy is similarly limited: it tests dependence on image conditioning, but it is not token-level visual contrastive decoding because the hosted backend does not expose the logits needed for that method. Finally, the cost/value table uses declared estimates rather than provider invoice records. These caveats define the exact scope of what the current rows can support.

10 Ethical Considerations

The manuscript avoids presenting diagnostic oracle rows as proposed-method evidence and avoids overstating unsupported hallucination-robustness claims. Future full-scale evaluation should report

unresolved rows, cost, and scoring caveats directly because hallucination mitigation claims can affect user trust in visual systems.

The ethical risk is mainly overclaiming. A training-free controller that appears more transparent can invite users to trust a visual answer because it has intermediate evidence, even when the evidence gate is not improving accuracy or coverage on the measured slice. This manuscript therefore keeps oracle rows out of method claims, reports the strong-comparator scope in the abstract and main table, and states that the HallusionBench and contrast results are protocol-limited. The same reporting discipline should apply to future extensions: failed rows, no-answer rows, estimated costs, and protocol restrictions should remain visible rather than being hidden behind a single aggregate score.

There is also a benchmark-use risk. Public hallucination and visual QA benchmarks are valuable because they make comparisons reproducible, but small visible slices can be overfit through repeated prompt changes. The present matrix is described as scoped for that reason. A later stronger claim should use larger or held-out slices, preserve the same baselines, and log accepted/rejected evidence traces so that a gain cannot be explained by cherry-picking tasks, weakening the comparator, or leaking diagnostic information into paper-facing rows.

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A Reproducibility Interface

The supplementary package provides a replay interface through row-level results, aggregate tables, paired comparison rows, benchmark provenance records, data-derived figures, and the method overview figure. Paper-facing reproduction should use the public benchmark families, the listed method conditions, and the same scoring policies; access credentials and machine-specific settings are outside the reader-facing protocol.

B Condition Definitions

Direct Answer is the simplest hosted visual response baseline. Evidence Prompt uses the same visual backend but asks for grounded evidence before the answer. Program Only emphasizes decomposition without the complete retrieval and memory path. RAG Only supplies retrieved context without the full evidence gate. No-Image Contrast Proxy removes image context as a diagnostic contrast and is not token-level visual contrastive decoding. EG-MPoT Realistic is the paper-facing controller that combines planning, retrieved evidence, gate decisions, memory, and answer composition. Oracle Diagnostic is excluded from proposed-method results because it uses dataset labels as evidence.

C Benchmark Family Context

The benchmark families were selected because each stresses a different reasoning regime. POPE is a direct object hallucination probe and is the cleanest place to test whether evidence gating suppresses unsupported object mentions (Li et al., 2023). HallusionBench mixes visual illusion with language hallucination and therefore requires care when narrowing the task to yes/no rows (Guan et al., 2023). A-OKVQA combines visual evidence with outside knowledge, but that same property makes answer extraction and option normalization fragile (Schwenk et al., 2022). MME is not executed in this matrix, but it remains a relevant example of broader LVLM evaluation coverage that a final paper could use as an additional public benchmark family (Fu et al., 2023).

The exemplar papers reinforce the same lesson. A hallucination paper needs dense benchmark context and cannot rely on a single aggregate score (Chen et al., 2024). A multimodal EMNLP paper can report constrained outcomes when the evidence warrants that framing (Khanuja et al., 2024). This

manuscript applies those structural lessons without importing their claims.

D Scoring Caveat Ledger

POPE is scored as a yes/no object hallucination task, but verbose model answers can interact with first-sentence coercion. HallusionBench is scored only as a constrained yes/no diagnostic slice, so its high baseline scores do not represent the official free-form setting. A-OKVQA uses direct-answer or multiple-choice exact matching. These caveats are benchmark-stage evidence limits, not appendix-only excuses.

E Artifact Inventory

The canonical expanded evidence consists of row-level scored results, aggregate result tables, paired comparisons, benchmark provenance records, data-derived figures, and the method overview figure. Headline numbers in the abstract, tables, and captions are tied to the same source rows.

F Raw Row Schema Summary

Each scored row records the benchmark family, task identifier, method condition, model-facing answer text, normalized score, coverage flag, error flag, scorer semantics, and whether the row is paper-facing or diagnostic-only. These row-level fields are needed because mean scores alone hide mechanism differences. For example, an A-OKVQA row can pass or fail because the extracted option matches the expected answer, but that exact-match outcome does not reveal whether retrieved evidence was necessary or merely correlated with the right choice.

The raw schema also records whether gold labels are exposed to the model. Paper-facing rows must keep that value false. Diagnostic oracle rows can expose labels only because they are excluded from proposed-method comparisons. This separation is important for anti-gaming: without it, a high oracle score could be mistaken for a usable system result.

G Scope Notes

- Broad EG-MPoT answer-accuracy superiority is outside the current claim; the supported contribution is the auditable evidence-admission operating point and its suitability map.
- HallusionBench protocol is constrained: official free-form judging is not represented.

Table 4: Appendix paired ledger over 180 paired comparison rows against strong non-oracle operating points.

Family	Comparator	Pairs	Delta	Count
A-OKVQA	Evidence Prompt	20	-0.200	4
HallusionBench	Direct Answer	20	-0.050	1
HallusionBench	Evidence Prompt	20	-0.050	1
HallusionBench	Program Only	20	-0.050	1
HallusionBench	RAG Only	20	-0.050	1
POPE	Program Only	30	-0.033	1
POPE	RAG Only	30	-0.033	1
POPE	Direct Answer	30	0.000	1

- Contrast proxy is not token-level VCD: the current condition removes image context but does not use decoder-logit visual contrast.
- External scale is bounded: the run covers 70 unique tasks and 490 scored method rows, not a large official leaderboard split.
- Provider invoice cost is missing: current cost/value rows are declared estimates only.

H Expanded Pair Ledger

The paired ledger is included to make the strong-comparator comparison auditable without relying on a single mean table. Each row compares EG-MPoT Realistic against a stronger or relevant non-oracle comparator on the same benchmark-family slice. The ledger confirms that the A-OKVQA gap is the largest observed gap and that HallusionBench and POPE gaps are smaller under the current protocol.

The pair counts also prevent a hidden denominator change. A-OKVQA and HallusionBench each have 20 paired tasks, while POPE has 30 paired tasks. The comparison is therefore not a cherry-picked subset of the expanded matrix; it is the family-level pairing used for the supplementary significance analysis. The bootstrap intervals are not repeated in every prose sentence, but the table records the signed direction and count that determine the paper’s scope.

I Literature Anchors for Extension

The method-side extension should remain anchored to tool-use and visual-program systems rather than becoming a generic prompt wrapper. ReAct motivates interleaving reasoning and actions (Yao et al., 2023), Toolformer motivates tool use as a

learned or prompted capability (Schick et al., 2023), MM-ReAct shows multimodal reasoning and action prompting (Yang et al., 2023), and ViperGPT shows the value and risk of programmatic visual inference (Suris et al., 2023). The hallucination-side extension should remain anchored to real visual contrast and retrieval evidence. Visual contrastive decoding is the relevant reference point for token-level contrast (Leng et al., 2023), retrieval visual contrastive decoding motivates a retrieval-aware variant (Lee and Song, 2025), and multimodal RAG benchmarks motivate reporting retrieval and planning separately (Li et al., 2024).

The expanded result says these ingredients should be measured directly. A strengthened paper should show whether the gate changes the answer distribution, whether retrieved evidence is actually used, and whether unresolved rows decrease after controlling for coverage. Those measurements would turn the current suitability map into a stronger hallucination-robustness claim (Huang et al., 2024).

J Operating Taxonomy Definitions

The expanded matrix motivates five operational categories for the next run rather than only a single success-rate metric. The first category is evidence over-filtering: the gate rejects or fails to admit enough evidence for an answerable item, reducing coverage or forcing a fallback response. The A-OKVQA coverage drop for EG-MPoT Realistic is the current example. The second category is evidence under-filtering: unsupported evidence enters memory and can shape the final answer. The present aggregate tables do not expose this mechanism directly, which is why a strengthened run should preserve accepted and rejected evidence traces.

The third category is comparator sensitivity. A method can look positive against Direct Answer while a stronger prompt, program, or retrieval baseline is preferable for answer-first accuracy. The expanded matrix shows this directly on A-OKVQA: the important comparator is Evidence Prompt at 0.850, not Direct Answer alone. The fourth category is protocol compression. Constrained HallusionBench yes/no scoring compresses several methods near 0.950, so the matrix can identify suitability under that protocol without making a complete official-benchmark statement. The fifth category is diagnostic leakage. Oracle evidence produces

useful scorer-ceiling information, but it must remain excluded from paper-facing proposed-method claims.

These definitions are intentionally tied to measured outcomes. Evidence over-filtering maps to the coverage columns in the main results matrix; comparator sensitivity maps to the strongest-baseline deltas in Figure 3; protocol compression maps to the HallusionBench scoring caveat; diagnostic leakage maps to the oracle rows excluded from proposed-method comparisons; and under-filtering is the remaining unmeasured mechanism that a method-extension study should instrument. This taxonomy is a concrete way to extend the current suitability map into a stronger next experiment.